Ashley Hu

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Education

Carnegie Mellon University May 2026, Pittsburgh, PA
B.S in Mechanical Engineering, Minor in CS *Major GPA: 3.75, Overall GPA: 3.57*
- Coursework: Mechanics II: 3D Design, Fluid Mechanics, Control Engineering, Mechanical Systems Experiment., Heat Transfer, Dynamics, Thermodynamics, Fundamentals of Programming, Intro to A.I

Work Experience

Joby Aviation, Battery Safety Engineering Intern May – Aug. 2026, San Carlos, CA
- Designing and fabricating 4 unique battery module text fixtures and a single-cell thermal sub-scale testing device for thermal runaway characterization and FAA certification using CATIA and metal machining
- Designing pre-charge circuit and 100V battery pack discharge station with voltage, current, and temperature tracking and remote E-stop

SpaceX, Production Operations Engineering Intern May – Aug. 2025, Bastrop, TX
- Created a hazardous waste tracking algorithm to improve production efficiency and designed a 2 story waste containment building in AutoCAD for entire facility, collaborating with EHS and Civil Engineering
- Lead the reconstruction of a production line with installations of 1.5 million dollars worth of equipment, water piping, and compressed air systems to production-critical areas
- Redesigned physical & electrical layout of Starlink dish testing farm, & led reconstruction with contractors

Tenaris, Mechanical Engineering Intern June – Aug. 2024, Koppel, PA
- Analyzed mechanical & electrical drawings of Continuous Casting Machine components & Torch Cutting Machine to create Machine Maintenance & Operation manuals for Maintenance Engineering team
- Automated data analysis & visualizations to track final steel bar quality with Power BI/Automate

Academic Experience

Carnegie Mellon Solar Racing | *President, Prev. Power Lead* Sept. 2022 – May 2026
- Designed & built five custom 2x5 ft solar arrays, performing electrical calculations & integrating the panels in a 36 V quad-battery system with a charge controller, motor controller, & full electrical wiring
- Design & assemble the motor controller, gearbox, & propulsion testing rig
- Lead team of 5 to place 2nd overall & 1st in endurance race at collegiate Solar Splash competition 2025
- Organize club meetings (board & general body), logistics for the two competitions we participate in

Carnegie Mellon Racing | *Low Voltage Battery Lead, Electrical Integration Member* Sept. – Jan. 2025
- Designed first custom low voltage battery for club using SolidWorks & manufactured battery, case, & mounts
- Performed FEA analysis & hand calculations to verify structural integrity of mounting brackets & casing
- Decreased weight by 1.5 kg & increased energy density by 75% compared to previous battery pack
- Designed & laser cut cell separators & attached thermal protection to assist in accumulator production

Project Experience

Automated Laundry Folder, Senior Capstone | *OnShape, KiCad, Arduino* Jan. 2026 – May 2026
- Created adjustable, automatic folder that detected clothing placement & shape in team of 5 with $800
- Sensing lead: Optimized sensor placement & noise reduction for item tracking, designed photoresistor matrix PCB, CADed sensor mounts, & wrote decision-making/microcontroller communication software
- CADed adjustable folding mechanism & clothing conveyor system in Onshape with Folding Mechanics Lead

Bear-Resistant Chair, Mechanics 2: 3D Design | *SolidWorks, Machining, Woodworking* March – May 2024
- Collaborated with team of 4 to design/manufacture chair able to bear 800 lb on uneven terrain
- Utilized iterative design process, hand calculations, & static analysis sims to optimize weight & strength

Automatic Pico Garden, CMU Build18 Contest | *SolidWorks, 3D Printing, Laser Cutting, Python* Jan. – Feb. 2024
- Collaborated with team of 5 to create self-watering & lighting desktop garden with interactive screen
- Designed & built watering system & garden enclosure, & coded display screen & moisture sensors

Technical Skills

Software: CATIA, Solidworks, Fusion 360, AutoCAD, Ansys Fluent & Discovery, Onshape, KiCAD, C, Python, Standard ML, Matlab, Arduino, SQL, NI LabVIEW, HTML, Power Automate, Power BI, Microsoft Office
Hardware/Tools: Manual Machines, 3D Printers, Laser Cutters, Power tools, Soldering
Languages: Spanish, Mandarin